

TRADING COST EQUATIONS

Our market impact and timing risk equations expressed in terms of percentage of trading volume POV are:

$$I_{bp}^* = \hat{a}_1 \cdot \left(\frac{X}{ADV} \right)^{\hat{a}_2} \cdot \sigma^{\hat{a}_3} \quad (7.1)$$

$$MI_{bp} = \hat{b}_1 \cdot I^* \cdot POV^{\hat{a}_4} + (1 - \hat{b}_1) \cdot I^* \quad (7.2)$$

$$TR_{bp} = \sigma \cdot \sqrt{\frac{1}{3} \cdot \frac{1}{250} \cdot \frac{X}{ADV} \cdot \frac{1 - POV}{POV}} \cdot 10^4 bp \quad (7.3)$$

where,

X = total shares to trade

ADV = average daily volume

σ = annualized volatility (expressed as a decimal, e.g., 0.20)

$POV = \frac{X}{X + V_t}$ = percentage of trading volume rate

V_t = expected market volume during trading period (excluding the order's shares X)

$\hat{a}_1, \hat{a}_2, \hat{a}_3, \hat{a}_4, \hat{b}_1$ = model parameters estimated via non-linear estimation techniques

Model Inputs

On the surface, the cost estimation process seems fairly straightforward, especially after having already estimated the model parameters. Investors simply need to enter their shares X and preferred POV execution strategy, and the model will determine cost estimates for these inputs.

Although, is the process really this simple and straightforward? Will the model provide accurate cost forecasts?

To answer these questions, let's take a closer look at our equations. Our transaction cost model actually consists of three different sets of input information:

1. User specified inputs: X, POV
2. Model parameters: $\hat{a}_1, \hat{a}_2, \hat{a}_3, \hat{a}_4, \hat{b}_1$
3. Explanatory factors: σ, ADV, V_t

The first set of input information is entered by the user and is based on the investment decision and the investor's urgency preference. In Chapter 5, Estimating I-Star Model Parameters, we provided non-linear regression techniques to estimate the parameters of the model and test the model's sensitivity. In Chapter 6, Price Volatility, we provided techniques to